Asteroid lightcurve inversion
Methods and reconstruction results

HAC–DTU project

Abstract

We reconstruct non-convex asteroid surfaces from multi-view rotational lightcurves using a spherical-harmonic octantoid mesh, differentiable image formation, empirical acquisition calibration, and a regularized staged Adam solver. We specify the forward model and optimization procedure, compare harmonic representations, and report saved degree-4 fits from a ten-asteroid experiment. Mesh and image comparisons expose the distinction between photometric agreement and geometric fidelity. Public references permit geometry evaluation for asteroids 1–3; interpretations of asteroids 4–10 are conditional on transferred calibration. All photometric errors are root-mean-square errors (RMSE) of normalized intensity lightcurves.

1 Problem, observations, and scope

For asteroid A, the simulated dataset contains $N = 360$ rotation samples and $C = 21$ independent cameras. Seven camera azimuths each provide horizontal, upper, and lower views. The duplicated horizontal channel is removed only after checking equality; the separate real-data loader can retain 28 channels. Public reference surfaces exist for asteroids 1–3; the shapes of asteroids 4–10 are hidden.

Let $\boldsymbol{p} \in \mathbb{R}^P$ be shape coefficients, F fixed triangle connectivity, and ϑ fixed acquisition parameters. The forward map is

$$\boldsymbol{p} \;\longmapsto\; (\boldsymbol{v}_i(\boldsymbol{p}), F) \;\longrightarrow\; I_{tj,u}(\boldsymbol{p}; \vartheta) \;\longmapsto\; B_{tj} \;\longrightarrow\; \widehat{y}_{tj}, \tag{1}$$

where t, j, and u index rotation, camera, and image pixel. Only the shape coefficients are fitted. The rotation axis is the world z axis; rotation samples and camera directions are supplied. The inverse solver fixes the global phase offset to zero after choosing a phase convention in calibration. It does not estimate a spin pole, rotation period, albedo map, or physical size.

2 Shape representation

2.1 Sphere mesh and harmonic fields

Let $\boldsymbol{u}_i \in S^2$ be fixed unit template directions. A standard subdivision-s icosphere has $20\,4^s$ faces and $10\,4^s + 2$ vertices. The exact 1,024-face template instead starts from a denser icosphere, uses quadric simplification, and reprojects vertices onto the unit sphere. The implementation checks the exact face count, closed topology, consistent outward orientation, and Euler characteristic two. The template is frozen and shared across the controlled fits.

Expand three scalar fields in real orthonormal spherical harmonics:

$$a(\boldsymbol{u}) = \sum_{\ell=0}^{L} \sum_{m=-\ell}^{\ell} a_{\ell m} Y_{\ell m}(\boldsymbol{u}), \qquad b(\boldsymbol{u}) = \sum_{\ell,m} b_{\ell m} Y_{\ell m}(\boldsymbol{u}), \qquad c(\boldsymbol{u}) = \sum_{\ell,m} c_{\ell m} Y_{\ell m}(\boldsymbol{u}). \tag{2}$$

For colatitude θ and longitude ϕ, the basis convention is

$$Y_{\ell m} = N_{\ell|m|} P_\ell^{|m|}(\cos\theta) \begin{cases} \sqrt{2}\sin(|m|\phi), & m < 0, \\ 1, & m = 0, \\ \sqrt{2}\cos(m\phi), & m > 0, \end{cases} \qquad N_{\ell k} = \sqrt{\frac{2\ell+1}{4\pi}\frac{(\ell-k)!}{(\ell+k)!}}. \tag{3}$$

The associated Legendre functions include the Condon–Shortley sign. Basis matrices are evaluated once at the template directions, so field evaluation during fitting is a matrix–vector product.

2.2 Octantoids and radial harmonics

The unsaturated octantoid mapping is

$$\boldsymbol{v}_i = r_0 \begin{bmatrix} u_{ix} e^{a_i} \\ u_{iy} e^{a_i + b_i} \\ u_{iz} e^{a_i + c_i} \end{bmatrix}, \qquad r_0 = 0.5, \tag{4}$$

following the octantoid approach to asteroid modelling [1, 2]. The implemented mapping smoothly bounds coordinate log-scales:

$$h_M(q) = M \tanh(q/M), \qquad \boldsymbol{v}_i = r_0 \boldsymbol{u}_i \odot \exp\left[h_M\left((a_i, a_i + b_i, a_i + c_i)^{\mathsf{T}}\right)\right], \qquad M = 3. \tag{5}$$

Exponentiation and h_M act componentwise. Positive coordinate multipliers preserve coordinate signs but do not enforce convexity or prevent self-intersection. The field a controls common radial deformation; b and c also allow displacement tangential to the initial sphere. This remains a restricted surface family with fixed spherical topology.

The uniform log-scale coefficient a_{00} is fixed to zero and omitted. The constants b_{00} and c_{00} remain free and permit axis-aligned ellipsoidal deformation. This fixes an explicit scale freedom; normalized photometry still supplies no physical length scale. With equal maximum degree in all fields,

$$P_{\mathrm{oct}} = 3(L+1)^2 - 1. \tag{6}$$

Setting $b = c = 0$ gives the implemented radial-harmonic subfamily,

$$\boldsymbol{v}_i = r_0 e^{h_M(a_i)} \boldsymbol{u}_i, \qquad P_{\mathrm{radial}} = (L+1)^2 - 1. \tag{7}$$

Thus octantoids L2/L4/L6 have 26/74/146 coefficients, while radial L6/L11 have 48/143. Harmonic degree controls deformation capacity; face count controls geometric discretization. Increasing one does not replace the other. All representation comparisons start from $\boldsymbol{p} = 0$, an identical sphere.

3 Forward image model

3.1 Rotation and observing geometry

For camera azimuth β_j and elevation α_j, the object-to-observer direction and light direction are

$$\boldsymbol{d}_j = \left(\cos(\beta_j - \pi)\cos\alpha_j, \sin(\beta_j - \pi)\cos\alpha_j, \sin\alpha_j\right)^{\mathsf{T}}, \qquad \boldsymbol{s} = (-1, 0, 0)^{\mathsf{T}}. \tag{8}$$

Horizontal elevation is zero. At azimuth zero, upper/lower elevations are $+21^\circ/-21^\circ$; at $45^\circ, 90^\circ, 135^\circ$ they are $+26^\circ/-24^\circ$; at $225^\circ, 270^\circ, 315^\circ$ they are $+24^\circ/-26^\circ$. These asymmetric elevations are retained in every fit.

At phase θ_t, vertices are rotated by $\boldsymbol{x}_{ti} = R_z(-\theta_t)\boldsymbol{v}_i$. The selected simulated-data convention is zero-based, $\theta_t = 2\pi t/360$ for $t = 0, \ldots, 359$; an alternative calibration candidate interprets frame labels 1–360 as degrees. Calibration uses one convention across all cameras, not separately fitted channel offsets.

Define $R = \max(10^{-6}, \max_i \|\boldsymbol{v}_i\|)$. A perspective observer has

$$\boldsymbol{e}_j = dR\boldsymbol{d}_j, \qquad \mathrm{FoV} = 2\arctan(1.20/d), \tag{9}$$

and looks at the origin with z up. Orthographic projection instead uses image-plane bounds $[-1.20R, 1.20R]$ and a constant viewing direction. The radius-relative camera/framing convention removes sensitivity to uniform mesh scale. Diagnostic ray-tracing comparisons can instead hold camera position and framing fixed in common reference coordinates.

3.2 Visibility, normals, and shading

The observer rasterizer retains up to four depth-ordered triangle fragments per pixel at 96×96 resolution with zero rasterization blur. Geometric triangles determine coverage and occlusion. Shading uses area-weighted vertex normals, interpolated barycentrically and renormalized at each fragment. Occluding faces remain in the rasterizer even when unlit.

A separate orthographic rasterization from $4R\boldsymbol{s}$ produces a 96×96 nearest-depth shadow map. If a fragment has light-view depth $z_L(\boldsymbol{x})$ and bilinearly sampled nearest depth $D_L(\boldsymbol{x})$, its soft solar visibility is

$$S(\boldsymbol{x}) = \mathrm{sigmoid}\left(\frac{D_L(\boldsymbol{x}) + 0.002R - z_L(\boldsymbol{x})}{\max(0.002R, 10^{-8})}\right). \tag{10}$$

A separately sampled validity mask treats missing shadow-map samples as lit. Observer depth ordering handles self-occlusion; the light depth test handles cast shadows. The finite light-camera position is a numerical construction: physical illumination is directional, with no inverse-square attenuation.

For unit shading normal $\boldsymbol{n}$, let

$$\mu_0 = \max(\boldsymbol{n} \cdot \boldsymbol{s}, 0), \quad \mu = \max(\boldsymbol{n} \cdot \boldsymbol{w}, 0), \quad \boldsymbol{w} = \frac{\boldsymbol{e}_j - \boldsymbol{x}}{\|\boldsymbol{e}_j - \boldsymbol{x}\|} \tag{11}$$

for perspective views; use $\boldsymbol{w} = \boldsymbol{d}_j$ in orthographic views. The calibrated Lambertian fraction $\xi \in [0, 1]$ mixes Lambert and Lommel–Seeliger radiance [3]:

$$L = S\left[\xi\mu_0 + (1 - \xi)\frac{\mu_0}{\max(\mu_0 + \mu, 10^{-3})}\right]. \tag{12}$$

There is no additional projected-area factor μ in this shader: projection already determines the pixel footprint. An equivalent orthographic surface integral would include geometric projected area and observer visibility explicitly. Multiplying by μ again in image space would double-count that effect.

PyTorch3D's depth-weighted softmax blend combines retained fragment radiances over black background, using blend parameters $\sigma_r = \gamma_r = 10^{-4}$ [4]. The model includes uniform material and direct illumination only. It does not include multiple scattering between surfaces, spatial albedo variations, or thermal emission. Rasterizer face selection remains discrete; the available derivatives describe a local surrogate rather than a globally smooth visibility map.

4 Image response and lightcurve extraction

The linear blended image I is transformed *before* either threshold rule:

$$E = \mathrm{clip}_{[0,1]}(T(gI)), \qquad T(x) = x \text{ or } \begin{cases} 12.92x, & x \le 0.0031308, \\ 1.055x^{1/2.4} - 0.055, & x > 0.0031308. \end{cases} \tag{13}$$

Here $g > 0$ is radiance gain and the second response is sRGB. It is an empirical candidate response, not proof of the original rendering software's color-management configuration. Gain can affect normalized curve shape because encoding, clipping, and thresholding are nonlinear.

4.1 Hard scoring and differentiable training

For validation and reporting, quantize to eight bits and apply the challenge intensity cutoff:

$$Q_u = \frac{\mathrm{round}(255E_u)}{255}, \qquad B_{tj}^{\mathrm{exact}} = \sum_u Q_{tj,u}\,\mathbf{1}[Q_{tj,u} > 0.1]. \tag{14}$$

Retained integer pixel values are 26–255. "Exact" describes this measurement reduction, not exact geometric optics: the underlying shadow map and fragment blend still use the preceding approximations.

Training omits quantization and substitutes a sigmoid gate with an exactly black background:

$$G_\kappa(E; \tau) = \mathrm{clip}_{[0,1]}\left[\frac{\mathrm{sigmoid}(\kappa(E - \tau)) - \mathrm{sigmoid}(-\kappa\tau)}{\max(1 - \mathrm{sigmoid}(-\kappa\tau), 10^{-8})}\right], \tag{15}$$

$$B_{tj}^{\mathrm{soft}} = \sum_u E_{tj,u} G_\kappa(E_{tj,u}; 0.1), \qquad \kappa = 8. \tag{16}$$

This surrogate supplies gradients; hard extraction selects checkpoints. Binary pixel-count acquisition is outside the intensity experiments reported here.

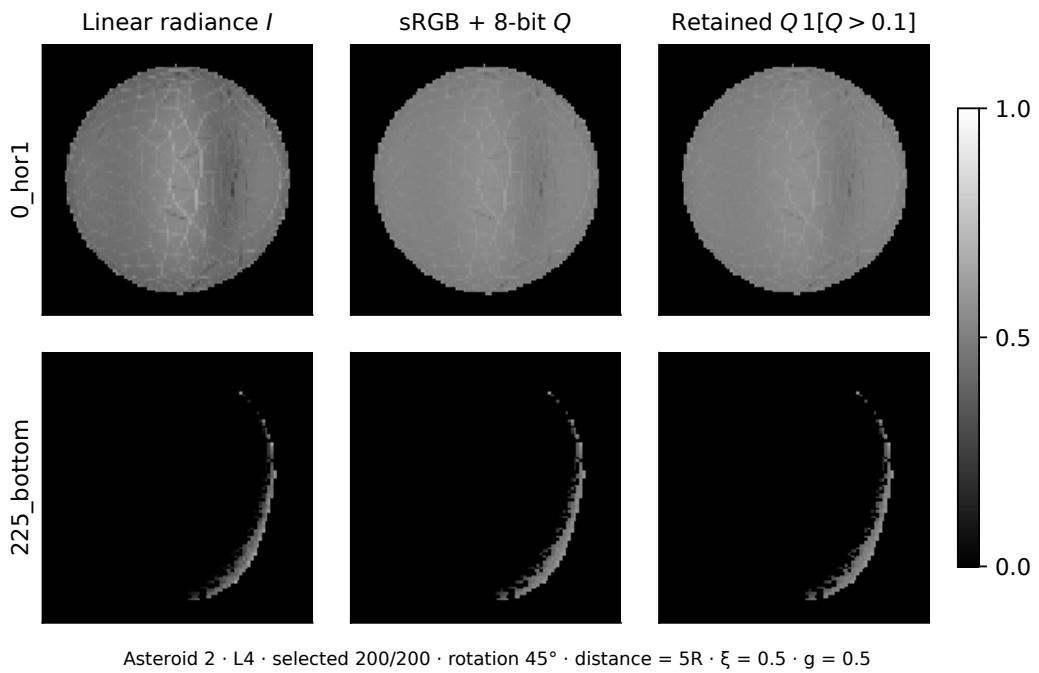


Figure 1: Image formation on the selected asteroid-2 L4 mesh (iteration 200 of 200), rendered on CPU at rotation 45°. Each panel contains the native 96×96 pixels enlarged without smoothing, on a common 0–1 gray scale. Left: linear radiance. Middle: sRGB response, gain 0.5, and eight-bit quantization. Right: retained intensity after the $Q > 0.1$ cutoff, not a binary silhouette. Perspective distance is $5R$, Lambert fraction $\xi = 0.5$, with a 96×96 shadow map and four fragments per pixel. Encoding brightens weak radiance before the cutoff; the `225_bottom` view retains a thin illuminated region.

4.2 Normalization and phase partitions

For a normalization set $\mathcal{T}$,

$$\bar{B}_j = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} B_{tj}, \qquad \widehat{y}_{tj} = \frac{B_{tj}}{\max(\bar{B}_j, 10^{-9})}. \tag{17}$$

Training normalizes both predictions and targets over all training rows. Render chunks never define separate normalization sets. Validation and reporting first render all 360 phases, normalize once over the full turn, and then select the scored rows. The supplied full-turn-normalized observations are not renormalized on validation subsets.

The experiments use disjoint zero-based index sets

$$\mathcal{T}_{\text{train}} = \{0, 5, \ldots, 355\}, \qquad \mathcal{T}_{\text{val}} = \{2, 7, \ldots, 357\}, \tag{18}$$

with 72 rows each and all 21 cameras. The other 216 phases do not select shape checkpoints. Denser phase coverage and full-turn scoring reduce normalization bias for narrow, high-contrast lightcurve peaks.

5 Empirical calibration

Calibration precedes inversion and uses the public bodies. Each reference is centered by its bounding box and scaled to z half-extent one. A 1,280-face radial proxy is constructed on a subdivision-3 icosphere: for each unit direction $\boldsymbol{u}_i$, intersect the ray $r\boldsymbol{u}_i$, $r > 0$, with the reference and retain the *outermost* intersection r_i. The proxy vertices are $r_i \boldsymbol{u}_i$ with the icosphere's triangle connectivity. A direction with no hit invalidates this construction. Figure 2 compares the actual asteroid-3 reference and its proxy.

One radius per direction cannot represent multiple radial layers or occluded inner surfaces. Finite angular sampling and interpolation across triangles can also round narrow depressions and the waist between lobes. The proxy is therefore a calibration approximation; it need not preserve all reference geometry. With the proxy fixed, calibration minimizes pooled intensity RMSE over acquisition parameters $\vartheta = (\text{projection}, d, \xi, T, g, \text{phase convention})$.

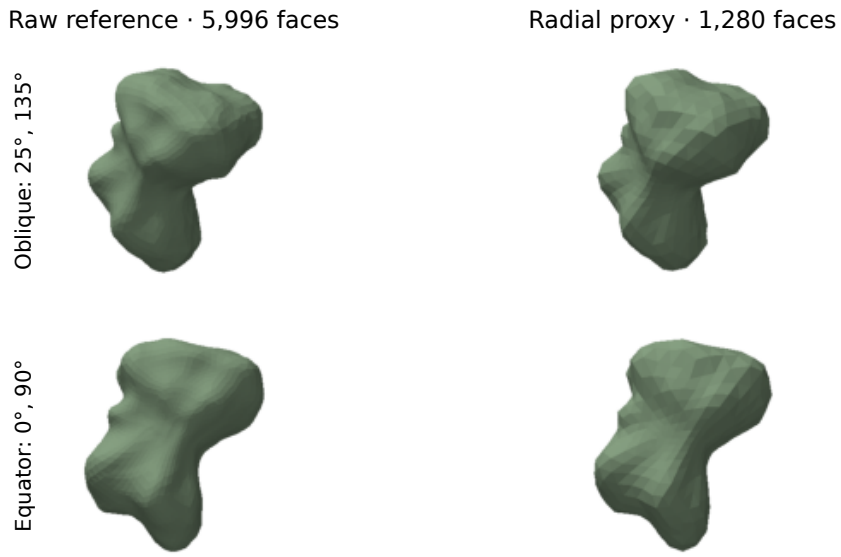


Figure 2: Asteroid 3: the raw 5,996-face reference and its 1,280-face radial proxy in matched orthographic views, at the same scale and orientation. The outermost-ray construction retains the broad two-lobed form but changes the waist and local surface detail. Flat triangle normals and common illustrative lighting make the tessellation visible; this is a geometry display, not an acquisition image.

The procedure screens 420 candidates per body: perspective distances $\{2.5, 3.5, 5, 8, 16, 32\}$ plus one orthographic case, mixtures $\{0, .25, .5, .75, 1\}$, two responses, gains $\{.5, 1, 2\}$, and two phase conventions. Screening uses 72 phases; the top three, a baseline candidate, and a shared-grid candidate are checked over all 360 phases. A winning finite-distance boundary triggers one outward extension. Because linear radiance is affine in ξ, Lambert and Lommel–Seeliger endpoint images can be mixed before response and thresholding; a direct-render check validates this acceleration.

Table 1: Public calibrations used for the L4 experiment. Scores are full-turn pooled RMSE of calibrated public proxies, not inverse-fit errors. All selected responses are sRGB with zero-based phase indexing.

Asteroid	Projection	d	ξ	g	Proxy RMSE
1	Perspective	5	1.00	1.0	0.016425
2	Perspective	5	0.50	0.5	0.161920
3	Orthographic	—	0.75	1.0	0.037210

Camera distances are effective acquisition parameters in radius units, not measured physical intrinsics. A shared calibration increased asteroid-2 and asteroid-3 proxy errors by 8.7% and 47.6% relative to separate settings in the calibration comparison, supporting per-body settings for public shapes.

Calibration uses public full-turn observations and shapes. Subsequent validation scores are therefore *conditional on calibration*, not end-to-end held-out reconstruction tests. Asteroids 4–10 use asteroid 3's calibration (orthographic projection, $\xi = 0.75$, sRGB, $g = 1$). This transfer is an assumption about acquisition; their hidden references cannot verify it or the inferred geometry.

6 Regularized inverse solver

6.1 Objective and staged updates

At zero-based update k, minimize

$$\mathcal{J}_k(\boldsymbol{p}) = \frac{1}{N_{\text{train}}C}\sum_{t,j}(\widehat{y}_{tj}^{\text{soft}}(\boldsymbol{p}) - y_{tj}^{\text{train}})^2 + \lambda s_k \mathcal{R}(\boldsymbol{p}), \qquad \mathcal{R}(\boldsymbol{p}) = \frac{1}{146}\sum_{r=1}^{P}\ell_r^3 p_r^2. \tag{19}$$

All samples and channels have equal weight. The denominator 146 is held fixed across the representation comparisons, rather than averaging over each model's own parameter count. Constants in b, c have zero spectral penalty. The prior suppresses high-degree deformation but is not a collision or convexity constraint.

The fixed optimizer settings used for the L4 experiment are

$$\lambda = 1.9962845 \times 10^{-3}, \quad \eta_{\text{init}} = 4.4673588 \times 10^{-3}, \quad \eta_{\text{base}} = 2 \times 10^{-3}, \quad \eta_{\min} = 10^{-5}. \tag{20}$$

With $T = 200$ updates, $E = 40$ escape updates, and $S = 40$ stabilization updates, the schedule is

$$\eta_k = \begin{cases} \eta_{\text{base}} + \dfrac{\eta_{\text{init}} - \eta_{\text{base}}}{2}[1 + \cos(\pi k/(E-1))], & 0 \le k < E, \\ \eta_{\min} + \dfrac{\eta_{\text{base}} - \eta_{\min}}{2}[1 + \cos(\pi(k-E)/(T-E-1))], & E \le k < T. \end{cases} \tag{21}$$

Escape masks gradients above degree two and sets $s_k = 0.25$. Refinement enables all coefficients with $s_k = 1$. During the last S updates, $s_k = 1 + [k - (T - S) + 1]/S$, ending at twice the base prior weight.

After the degree mask, clip the gradient to Euclidean norm 10 and apply Adam [5] with $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\epsilon = 10^{-8}$:

$$\boldsymbol{m}_k = \beta_1 \boldsymbol{m}_{k-1} + (1-\beta_1)\boldsymbol{g}_k, \qquad \boldsymbol{v}_k = \beta_2 \boldsymbol{v}_{k-1} + (1-\beta_2)\boldsymbol{g}_k^{\odot 2}, \tag{22}$$

$$\boldsymbol{p}_{k+1} = \boldsymbol{p}_k - \eta_k \frac{\widehat{\boldsymbol{m}}_k}{\sqrt{\widehat{\boldsymbol{v}}_k} + \epsilon}. \tag{23}$$

Hats denote bias-corrected moments. Moments persist across stages; there is no weight decay. Parameter projection is the identity. During fitting, finite-vertex checks detect numerical failure; post-fit mesh audits additionally inspect degeneracy and transverse self-intersections.

6.2 Memory-bounded differentiation

Training brightness depends on the mean over all training phases. Retaining every renderer graph would be unnecessarily expensive. The implemented two-pass scheme first renders all raw brightness in six-frame blocks without renderer graphs, then differentiates the small 72×21 brightness array. Away from the mean floor, writing $D_{tj} = \partial \mathcal{L}_{\text{data}}/\partial \widehat{y}_{tj}$ gives

$$\frac{\partial \mathcal{L}_{\text{data}}}{\partial B_{kj}} = \frac{D_{kj}}{\bar{B}_j} - \frac{\sum_t D_{tj}B_{tj}}{N_{\text{train}}\bar{B}_j^2}. \tag{24}$$

Each block is then rerendered with gradients, and these brightness derivatives are backpropagated. Add the prior once and take one optimizer step after all blocks. This is the chain rule for the complete-set normalization, not independent mini-batch optimization. It trades an extra forward pass for bounded graph memory. Algebraic and CUDA checks compare it with unchunked differentiation.

6.3 Validation selection and reported errors

Hard extraction evaluates the sphere and every 20 updates. The selected state minimizes

$$\text{RMSE}_{\text{val}} = \sqrt{\frac{1}{72C}\sum_{t\in\mathcal{T}_{\text{val}},j}(\widehat{y}_{tj}^{\text{exact}} - y_{tj})^2}. \tag{25}$$

The full-turn pooled RMSE in all result tables instead uses all 360 phases and all 21 cameras with equal weight. It is a descriptive fit error, including both training and validation phases. Selection by validation can therefore produce a slightly higher full-turn error than the terminal state. The tables distinguish selected iteration from update budget.

7 Experimental settings and degree comparison

All fits use seed zero, a radius-0.5 sphere with 1,024 faces, 96×96 images and shadow maps, 21 cameras, the 72/72 phase split, six-frame training blocks, fixed calibration, and the prior denominator 146. Table 2 distinguishes the optimization budgets.

Table 2: Experiments represented in the results. E/R/S denotes escape, refinement, and stabilization updates; the degree comparison also applies plateau stopping. No run is implied by a missing table entry.

Experiment	Models	Update budget	E/R/S
Ten-asteroid experiment	Octantoid L4	200	40/120/40
Asteroid-3 screening	Radial L6, L11; octantoid L6	120	12/96/12
Asteroid-1 degree comparison	Octantoid L2, L4, L6	Cap 360	40/280/40*

*Nominal schedule before stopping. After at least 200 updates and 100 without meaningful validation improvement, 40 confirmation updates use a learning-rate ceiling reduced by factor 0.2. Improvement must exceed $\max(10^{-4}, 0.005\ \mathrm{RMSE}_{\mathrm{reference}})$. This is a stopping rule, not a certificate of optimality.

Table 3: Asteroid-1 degree comparison. Scores are full-turn pooled RMSE; stop iteration is the number of updates actually used.

Model	Selected iteration	Stop iteration	Selected RMSE	Terminal RMSE
L2	160	300	0.034558	0.035676
L4	180	320	0.027107	0.028562
L6	40	240	0.036864	0.124039

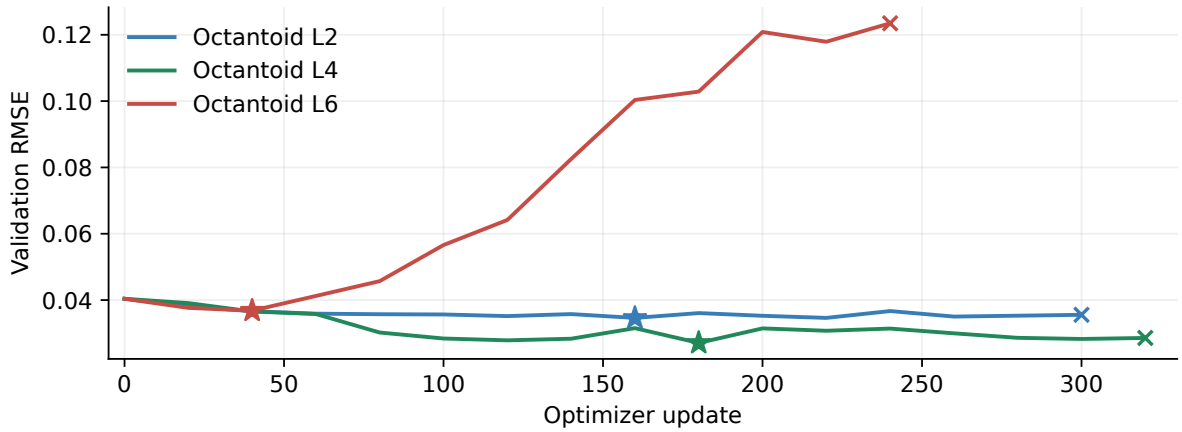


Figure 3: Exact-extraction validation histories for the asteroid-1 degree comparison. Stars mark selected states; crosses mark terminal states. Histories end at their measured stopping iterations.

L4 has the lowest selected photometric error among these three asteroid-1 fits. L6 deteriorates after its early selected state while remaining numerically finite. The short asteroid-3 L6 fit below demonstrates that this behavior depends on the object and optimization recipe. These comparisons do not identify the best attainable performance of each degree under separate tuning.

8 L4 results and qualitative reconstruction

The relative photometric improvement is $100(\mathrm{RMSE}_{\mathrm{initial}} - \mathrm{RMSE}_{\mathrm{selected}})/\,\mathrm{RMSE}_{\mathrm{initial}}$.

Table 4: Ten-asteroid L4 experiment: identical 1,024-face sphere starts, 74 coefficients, and 200 updates per reported fit. RMSE is pooled over the full turn and all cameras. Percentage improvement uses unrounded scores. — means "no result reported." † denotes calibration transferred from asteroid 3.

Asteroid	Selected iteration	Initial RMSE	Selected RMSE	Improvement
1	140	0.0403	0.0277	31.1%
2	200	0.6989	0.3669	47.5%
3	200	0.1544	0.0870	43.7%
4†	100	0.2348	0.0675	71.3%
5†	200	0.1100	0.0934	15.1%
6†	200	0.4598	0.1517	67.0%
7†	60	0.2979	0.2466	17.2%
8†	—	—	—	—
9†	—	—	—	—
10†	—	—	—	—

8.1 Asteroid 2: broad outline and residual discrepancy

The 200-update L4 fit reduces full-turn RMSE from 0.6989 to 0.3669 (47.5%). Figure 4 shows the selected state, not a smoothed or reference-adjusted replacement. Its approximately square outline is visible in the north view. Rounded edges and the absence of the raw reference's detailed cuts remain clear in the matched views. A plausible broad outline is weaker evidence than recovering the actual surface.

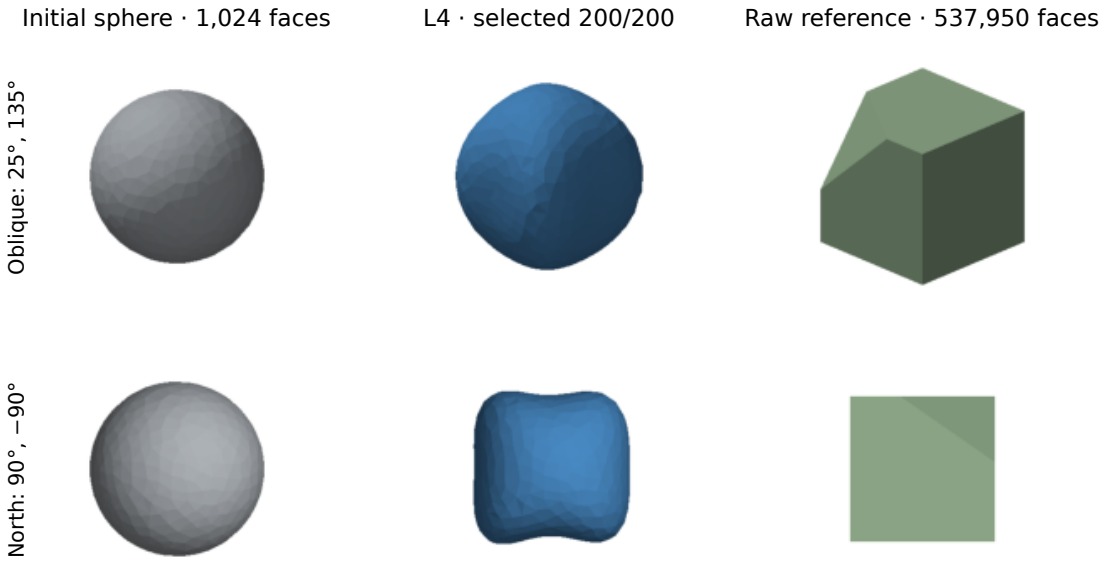


Figure 4: Asteroid 2: initial sphere, selected L4 reconstruction at iteration 200 of a 200-update fit, and the raw 537,950-face reference. Initial and selected meshes each have 1,024 faces. Display scales are independently aligned to the raw reference using the radial-log criterion below; no rotation is fitted for the display. Matched camera orientation and common framing reveal the broad square outline and the missing geometric detail.

8.2 Asteroid 3: photometry and geometric fidelity

The separate 120-update comparison in Table 5 uses the same sphere and calibration across representations. The selected octantoid L6 state has the lowest photometric error, reducing the initial 0.154377 to 0.061364. It differs from asteroid 3's *L4, 200-update* result in Table 4; these are different models and budgets.

Table 5: Asteroid-3 representation screening: 120 updates and common calibration, mesh, and prior denominator. Scores are full-turn pooled RMSE.

Representation	Parameters	Selected iteration	Selected RMSE	Terminal RMSE
Radial harmonics L6	48	80	0.069387	0.071523
Radial harmonics L11	143	40	0.066411	0.229551
Octantoid L6	146	120	0.061364	0.061364

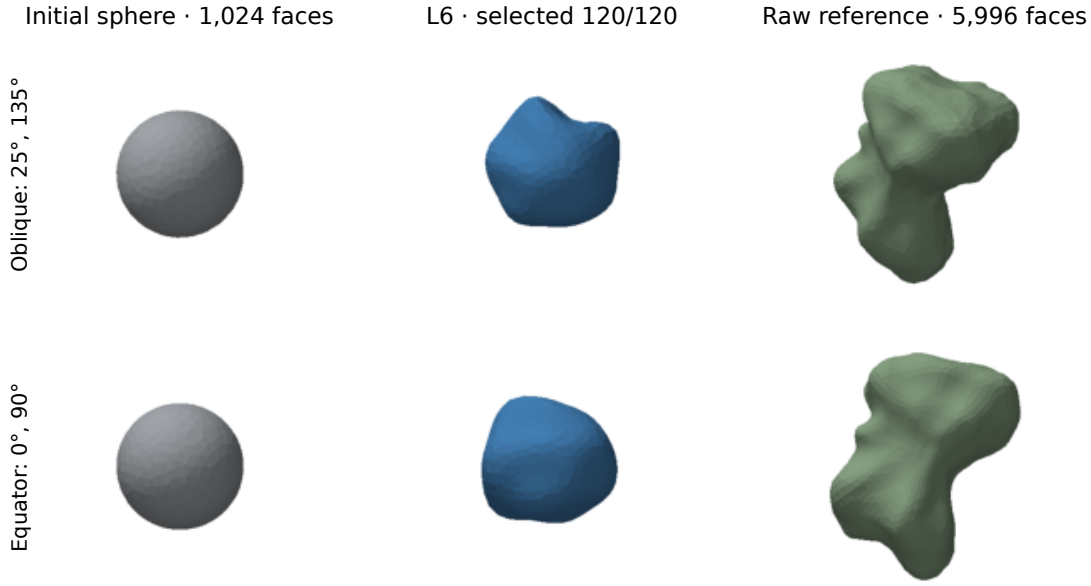


Figure 5: Asteroid 3: initial sphere, selected octantoid L6 at iteration 120 of a 120-update fit, and the raw 5,996-face reference. Matched views and scale-only alignment expose a broad two-lobed tendency but a substantial discrepancy in lobe proportions, neck shape, and overall extent. The reference here is the raw surface, not the 1,280-face proxy in Figure 2. The 120-update budget differs from the 200-update L4 examples for asteroids 2 and 6.

For public bodies, a geometry diagnostic samples 4,096 common Fibonacci directions and compares outer radii $r_q^{\mathrm{fit}}, r_q^{\mathrm{ref}}$ after aligning a global scale only:

$$\log s^* = \frac{1}{Q}\sum_q \log \frac{r_q^{\mathrm{ref}}}{r_q^{\mathrm{fit}}}, \qquad E_{\mathrm{geom}} = \sqrt{\frac{1}{Q}\sum_q \left[\log \frac{s^* r_q^{\mathrm{fit}}}{r_q^{\mathrm{ref}}}\right]^2}. \tag{26}$$

This measures the outer radial surface, not hidden cavities. The selected asteroid-2 L4 geometry error is 0.1064 (initial 0.1193). For asteroid 3, selected L6 octantoid error is 0.4930, compared with 0.4739 for radial L6, despite the octantoid's lower photometric error. Figure 5 makes this limitation visible. Neither the surface prior nor photometric selection guarantees geometric accuracy.

9 Asteroid 6 and interpretation limits

Asteroid 6's selected L4 state is iteration 200 of a 200-update fit. Its full-turn RMSE improves from 0.4598 to 0.1517 (67.0%). The complementary views in Figure 6 suggest a box-like form with a crater-like depression. This is an **unverified interpretation**, conditional on the calibration transferred from asteroid 3. There is no public reference surface for asteroid 6. Photometric agreement does not verify the depression, its depth, or the overall geometry.

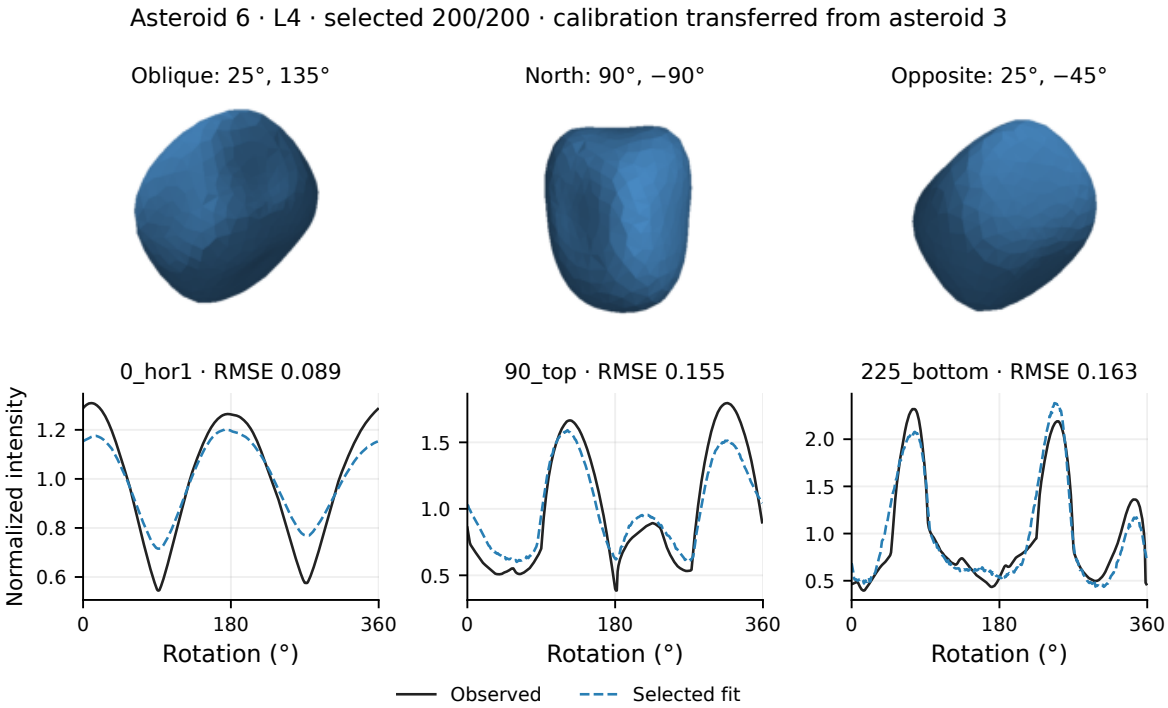


Figure 6: Asteroid 6: complementary geometry views of the same selected L4 mesh (200/200 updates), with observed and fitted full-turn lightcurves for a horizontal, an upper, and a lower channel. These channels sample different viewing geometries; the table score uses all 21. Curves are reproduced directly from the saved selected fit. The mesh views use common illustrative lighting and framing. A box-like form and crater-like depression are unverified interpretations under transferred calibration.

Normalized photometry is non-unique: different shapes and acquisition settings can yield similar disk-integrated curves. The finite harmonic basis, fixed mesh connectivity, proxy calibration, approximate shadows, and difference between soft training and hard scoring introduce further model dependence. Fixed spherical topology excludes disconnected bodies; positive coordinate multipliers do not exclude self-intersection. Mesh validity audits detect numerical or geometric defects, but they cannot establish correspondence to a hidden reference. No physical size, albedo map, spin pole, or period is inferred by these fits.

Figures and tables reproduce saved fits without changing experiment settings. The source bundle contains generated figures, native image arrays, table fragments, and input hashes. Claims are restricted to the reported measurements and budgets; missing entries supply no evidence about fit quality or geometry.

References

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